

Journal – Week 1

Why did I choose AI Engineering?

I have always been drawn to technology that seems to think, and artificial intelligence sits right at the center of that curiosity. What pulled me toward AI Engineering specifically was my interest in problem-solving. I enjoy taking a messy, real-world problem and breaking it down into something a system can actually handle, whether that means automating a repetitive task or building an application that responds intelligently to a user.

AI Engineering appealed to me because it is not purely theoretical. It combines the practical discipline of software development, writing clean code, structuring projects, working with APIs, with the rapidly evolving world of AI technologies like large language models. That mix of building real, usable software while also working with cutting-edge models felt like the right place for me to grow.

What type of AI products do I want to build?

I am most interested in building AI-powered web applications and AI assistants that help people get things done faster. Chatbots, productivity tools, and automation systems are especially exciting to me because they solve everyday problems, saving someone time, reducing manual work, or making information easier to access.

Longer term, I want to build AI agents that can carry out multi-step tasks on their own, and business solutions that bring real value to organizations rather than just being technical demonstrations. These products matter to me because they are practical. I would rather build something that genuinely helps someone in their workday than something that is only impressive on the surface.

What was my biggest learning this week?

This week was dense with new concepts. I was introduced to large language models (LLMs) and how they generate responses, and I started learning prompt engineering, how the way you phrase and structure a prompt directly affects the quality of the output. I also got my first real exposure to AI agents and tool calling, which was a bit of a turning point for me. Seeing how a model can be given tools and decide when to use them made AI feel far less like a black box.

I also spent time on memory in AI systems, working with APIs, exploring Hugging Face, and organizing code on GitHub. Individually, none of these topics felt overwhelming, but together they gave me a clearer picture of the full AI application development workflow, from a raw idea to a working prototype. It changed my understanding of modern AI systems from "a chatbot that answers questions" to a set of components, models, tools, memory, and orchestration, that can be combined deliberately to build something useful.

What challenges did I face?

- Understanding AI Engineering concepts, since many terms and workflows were new to me.
- Learning prompt engineering techniques and figuring out why some prompts worked better than others.
- Configuring APIs and environment variables correctly, especially keeping keys secure.
- Running into Hugging Face model access issues and API errors that took time to interpret.
- Debugging Python code when scripts failed silently or threw unclear error messages.
- Organizing project files and my GitHub repository in a way that stayed clean as the project grew.

How did I solve them?

Whenever I got stuck, I went back to the official documentation first, since most answers were already there if I read carefully. I also learned by experimenting directly, testing small changes and observing the results rather than only reading about them. When code broke, I debugged systematically, isolating the problem line by line instead of guessing.

I was not afraid to ask questions when I genuinely could not find an answer on my own. For prompt engineering specifically, I improved through iteration, rewriting a prompt several times until the output matched what I actually needed. Overall, I tried to treat mistakes as normal parts of the process rather than setbacks, and each error taught me something I did not know before.

What are my goals for Week 2?

- Learn AI agents in greater depth, including how they plan and use tools.
- Build a complete, working AI-powered application from start to finish.
- Improve my core Python programming skills.
- Explore LangChain and other agent frameworks.
- Get more practice with API integration in real projects.
- Create cleaner, more organized project structures.
- Improve my GitHub portfolio so it reflects my progress clearly.
- Continue building on what I learned about prompt engineering this week.

Conclusion

Looking back on this first week, I feel like I have a much clearer starting point than when I began. There is still a lot I do not know, but that no longer feels intimidating, it feels like the normal shape of learning something new and worthwhile. I am genuinely excited to keep building, keep making mistakes, and keep improving as I work toward becoming a skilled AI Engineer.